

Elements of Bayesian statistical analysis

Partly covered by chapter 11 in the textbook, and more extensively by these notes.

Most likely, up to now you have mostly encountered frequentist (or classical) statistics in your studies. These notes consider a somewhat different take on statistical estimation and characterization of uncertainty in general.

Before we start, let's introduce some ("grown up") notation:

Instead of the conventional notation for densities (or mass functions), e.g. $f_X(x)$, $g_Y(y)$, we use the notation $p(x)$ to denote the density of random variable x , $p(y|x)$ is the density of random variable $y|x$ and so on (notice, no distinction between upper and lower case letters). I.e. "read out of the argument" which density we are talking about.

Further, we let vectors be denoted by bold-face, i.e. $\mathbf{x}$ is a vector, x is a scalar.

Overview

Typical Bayesian statistics workflow:

- Before observing data, state what is known about the phenomenon you are interested using a *prior* distribution
- Once data are observed, update the prior using Bayes law to obtain the *posterior* distribution. The posterior distribution gives a coherent way of expressing knowledge/uncertainty about the phenomenon of interest after seeing data.
- Posterior distribution may be interesting in it self, or may be used, e.g. for predicting future outcomes.

Both prior and posterior are handled mathematically as random variables with densities etc (unlike classical statistics where we are interested in an unknown, but fixed quantity). Small variance = rather certain, large variance = highly uncertain, and so on.

The prior

Unlike classical statistics, the parameters $\boldsymbol{\theta}$ of a model are allowed to have a distribution, even before any data are observed. We use the random variable machinery as a coherent way of expressing the uncertainty about the parameters.

$$p(\boldsymbol{\theta})$$

is known as the *prior distribution*. It encapsulates what is known about the parameter before any data is collected.

Examples of priors:

- θ = Temperature at a given position at some timepoint in the future. Prior centered around historical mean temperature at position and time of year, and with variance similar to observed variance at position and time of year.
- θ = Geology of some unexplored oilfield. Prior should encapsulate general geological knowledge specific to the region.
- θ = Parameters of the SEIR Covid-19 model. Prior should reflect current best knowledge about Covid-19 and similar Corona-viruses.
- θ = Shape of an object. Prior should reflect a shape distribution learned from a training data set.

Example: θ = Temperature at UiS on Jan 1st next year measured in Celsius. Common knowledge:

- All temperatures are > -273.15 C
- Temperatures in south-western Norway in winter vary between say -10 and 10 Celsius.
- Reasonable prior e.g. $\theta \sim N(0, 10^2)$

The likelihood function

The likelihood function $L(\boldsymbol{\theta})$ plays a similar role as in classical statistics. The likelihood function is nothing more than the joint probability density/mass function of the data $\mathbf{y}$ *conditionally* on the parameter $\boldsymbol{\theta}$, i.e.

$$L(\boldsymbol{\theta}) = p(\mathbf{y}|\boldsymbol{\theta})$$

The likelihood function describes how measurements/data $\mathbf{y}$ are distributed, conditionally on the parameter $\boldsymbol{\theta}$.

cont. Example: θ = Temperature at UiS on Jan 1st:

Data: y =temperature reading from yr.no for Stavanger on Jan 1st. Statistical model e.g.

$$y|\theta \sim N(\theta, \sigma^2), L(\theta) = \frac{\exp\left(-\frac{(y-\theta)^2}{2\sigma^2}\right)}{\sqrt{2\pi\sigma^2}}$$

where σ reflect both standard measurement errors (thermometer, rounding) and the fact that we are using observations not exactly at UiS (elevation of "Stavanger" different from UiS).

The posterior distribution

Using Bayes theorem, the posterior distribution $p(\boldsymbol{\theta}|\mathbf{y})$ (i.e. distribution of $\boldsymbol{\theta}$ after seeing data) obtains as

$$p(\boldsymbol{\theta}|\mathbf{y}) = \frac{p(\mathbf{y}, \boldsymbol{\theta})}{\int p(\mathbf{y}, \boldsymbol{\theta}) d\boldsymbol{\theta}}$$

Notice 1: the numerator $p(\mathbf{y}, \boldsymbol{\theta}) = p(\mathbf{y}|\boldsymbol{\theta})p(\boldsymbol{\theta}) = L(\boldsymbol{\theta})p(\boldsymbol{\theta})$.

Notice 2: The shape, with respect to $\boldsymbol{\theta}$ of $p(\boldsymbol{\theta}|\mathbf{y})$ is the same as that of $L(\boldsymbol{\theta})p(\boldsymbol{\theta})$.

Hence we write

$$p(\boldsymbol{\theta}|\mathbf{y}) \propto L(\boldsymbol{\theta})p(\boldsymbol{\theta})$$

Notice 3: denominator = $p(\mathbf{y})$ = "evidence"

cont. Example: θ = Temperature at UiS on Jan 1st:

Joint distribution of parameter and data:

$$p(\mathbf{y}, \boldsymbol{\theta}) = \frac{\exp\left(-\frac{(\mathbf{y}-\boldsymbol{\theta})^2}{2\sigma^2}\right)}{\sqrt{2\pi\sigma^2}} \frac{\exp\left(-\frac{\boldsymbol{\theta}^2}{200}\right)}{\sqrt{200\pi}}$$

In this case $p(\mathbf{y}, \boldsymbol{\theta})$ is a bivariate normal distribution, i.e.

$$\begin{pmatrix} \mathbf{y} \\ \boldsymbol{\theta} \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 100 + \sigma^2 & 100 \\ 100 & 100 \end{pmatrix} \right)$$

The evidence ($\mathbf{y}$ -marginal): $p(\mathbf{y}) = \int p(\mathbf{y}, \boldsymbol{\theta}) d\boldsymbol{\theta}$
:

$$\mathbf{y} \sim N(0, 100 + \sigma^2)$$

Posterior:

$$\boldsymbol{\theta}|\mathbf{y} \sim N \left(\frac{100\mathbf{y}}{100 + \sigma^2}, \frac{100\sigma^2}{100 + \sigma^2} \right)$$

Recall

$$\boldsymbol{\theta}|\mathbf{y} \sim N\left(\frac{100\mathbf{y}}{100 + \sigma^2}, \frac{100\sigma^2}{100 + \sigma^2}\right)$$

- σ "small" (measurements highly informative) gives $E(\boldsymbol{\theta}|\mathbf{y}) \approx \mathbf{y}$ and $Var(\boldsymbol{\theta}|\mathbf{y}) \approx \sigma^2$. Almost all information with respect to $\boldsymbol{\theta}$ "comes from" the measurement.
- σ "large" (measurements very noisy) gives $E(\boldsymbol{\theta}|\mathbf{y}) \approx 0$ (prior mean) and $Var(\boldsymbol{\theta}|\mathbf{y}) \approx 100$ (prior variance). Almost all information with respect to $\boldsymbol{\theta}$ "comes from" prior.

Using the posterior

Probability of slippery hill up to UiS after seeing today's observation for Stavanger :

$$P(\boldsymbol{\theta} < 0) = \frac{\text{pnorm}(0, 100*y/(100+\text{sigmaSq}), \sqrt{100*\text{sigmaSq}/(100+\text{sigmaSq}))}}{1}$$

Find A, B so that $P(A < \boldsymbol{\theta} < B) = 0.95$:

$$\frac{\text{qnorm}(c(0.025, 0.975), 100*y/(100+\text{sigmaSq}), \sqrt{100*\text{sigmaSq}/(100+\text{sigmaSq}))} - \text{qnorm}(c(0.025, 0.975), 100*y/(100+\text{sigmaSq}), \sqrt{100*\text{sigmaSq}/(100+\text{sigmaSq}))}}{2}$$

A and B may be used to inform choice of jacket.

Notice, no "in repeated" sampling statement as in classical statistic.

The posterior convey the complete information we have about $\boldsymbol{\theta}$

Using the posterior

Classical inference analogs

Answering question of the types

- Probability of positive $\boldsymbol{\theta}$: $P(\boldsymbol{\theta} > 0) = \int_0^\infty p(\boldsymbol{\theta}|\mathbf{y})d\boldsymbol{\theta}$.
- $\boldsymbol{\theta}$ is with 95 % certainty between A and B (A and B can simply be taken as the 2.5% and 97.5% quantiles of $p(\boldsymbol{\theta}|\mathbf{y})$ respectively.)

Uncertainty in derived quantities

Supposed you are interested in temperature in Kelvin, say $\eta = \boldsymbol{\theta} + 273.15$ rather than Celsius, than

$$\eta|\mathbf{y} \sim N \left(273.15 + \frac{100\mathbf{y}}{100 + \sigma^2}, \frac{100\sigma^2}{100 + \sigma^2} \right).$$

Similar calculations can be done for any non-linear transformation of the parameter using Monte Carlo output to be discussed soon.

Much more...

Sequential inference,

Summary:

- Bayesian inference gives a principled way of combining information from prior beliefs and data using Bayes rule.
- The use of random variable machinery also for the parameter gives a coherent way of characterizing any form of uncertainty.
- Bayesian inference does not rely on the "repeated sampling" assumption of classical statistic. I.e. in Bayesian inference it makes sense to talk about e.g. $P(\boldsymbol{\theta} > 0)$ based on a single data set etc.
- BUT finding the exact posterior is in general difficult - we will spend the rest of this course discussing Monte Carlo methods for drawing random samples from the posterior distribution.